

Q.1: SANDSTONE BED INTERPRETATION

QUESTION:

A normal graded bed of sandstone is noticed with flute cast at its base and internally made up of **massive, plane-laminated and ripple-laminated subdivisions**, in order of superposition. The sandstone bed is interpreted as a product of:

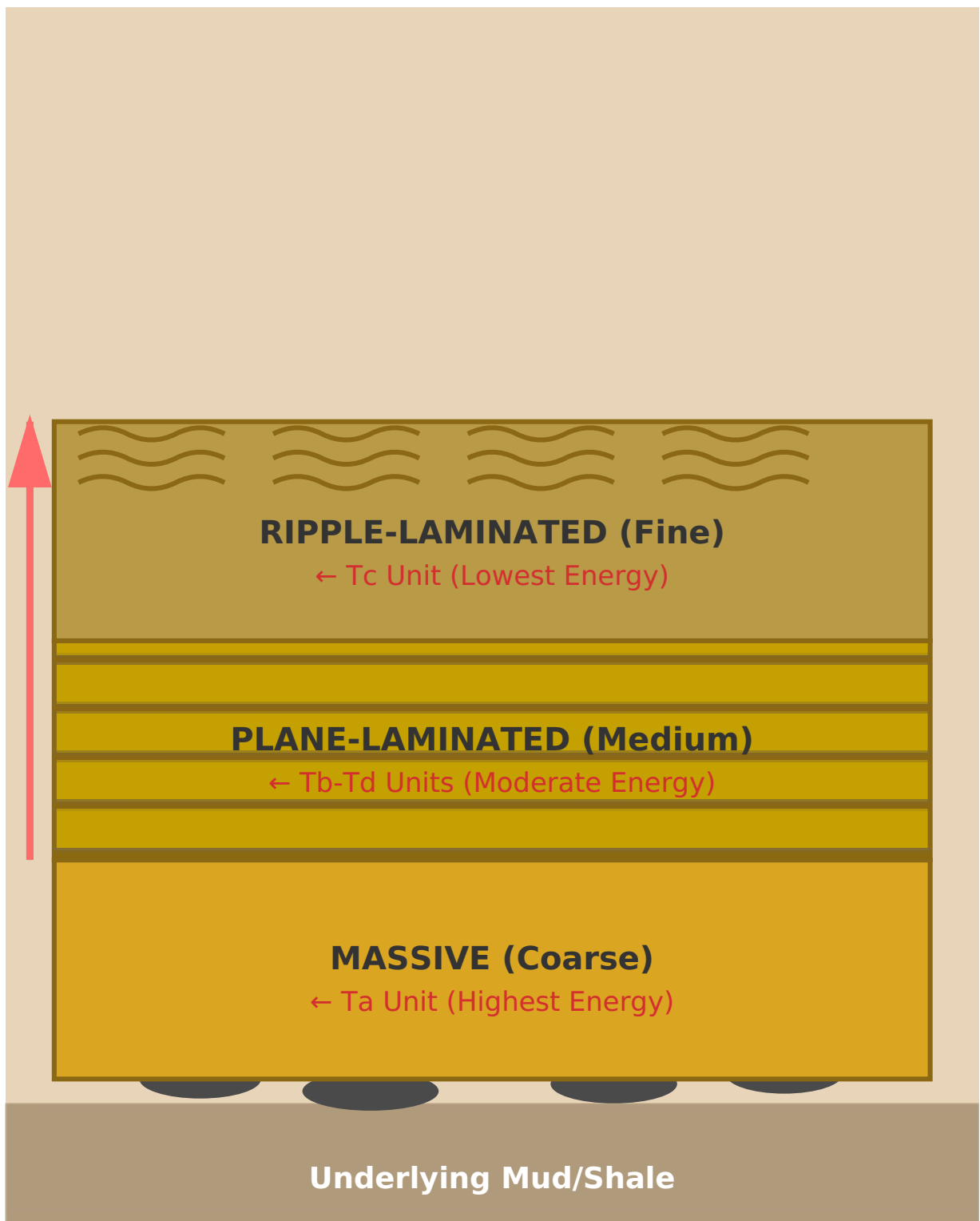
- (a) Debris flow
- (b) Grain flow
- (c) Turbidity current
- (d) Contour current

✓ **ANSWER: (c) TURBIDITY CURRENT**

 **SOURCE:** "Principles of Sedimentology and Stratigraphy" (4th Edition)
By Sam Boggs, Jr. | Pearson Prentice Hall, 2006 | ISBN: 0-13-154728-3

1 What Your Question Describes

The Sandstone Bed Structure



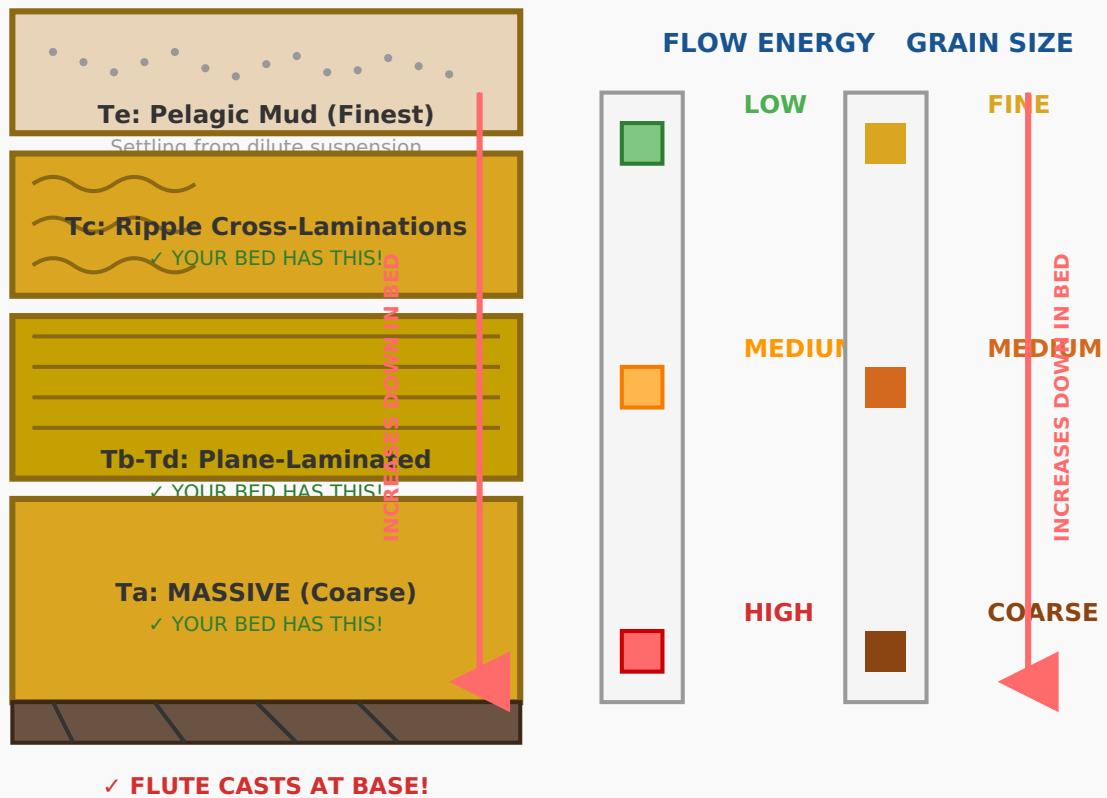
Your Sandstone Bed: Flute casts at base + Normal grading (coarse → fine) + Progressive structures = TURBIDITE

🔍 Key Observations in Your Bed:

- ✓ **Flute casts at base** - Sole marks indicating high-energy turbulent scour
- ✓ **Massive division (bottom)** - Rapid deposition from dense flow
- ✓ **Plane-laminated (middle)** - Traction transport as flow weakens
- ✓ **Ripple-laminated (top)** - Low-energy ripple formation
- ✓ **Normal grading** - Coarse at base, fine at top (suspension settling)

2 The Bouma Sequence: 5-Unit Framework

How Your Bed Matches the Bouma Sequence



The Bouma Sequence: Five units from high to low energy, exactly matching your bed description

3 Analysis of ALL FOUR Options

Comparison of Deposit Types

× (a) DEBRIS FLOW

What it produces:

- Chaotic, unsorted deposits
- NO internal layering
- Random clast orientation
- Poor to reverse grading

✗ **YOUR BED HAS: Organized layers, normal grading, parallel laminations**

✗ (b) GRAIN FLOW**What it produces:**

- Massive, structureless
- Little/NO laminations
- Very thin units (<5cm)
- Possible reverse grading

✗ **YOUR BED HAS: Multiple laminations, ripple patterns, good thickness**

✓ (c) TURBIDITY CURRENT**What it produces:**

- Organized Bouma sequence
- Massive → laminated → rippled
- NORMAL grading (coarse→fine)
- Flute casts & sole marks

✓ **PERFECT MATCH WITH YOUR BED!**

✗ (d) CONTOUR CURRENT**What it produces:**

- Laminated mud & silt deposits
- Ripple marks (NOT flute casts)
- NO normal grading
- Flows parallel to contours (not downslope)

✗ **YOUR BED HAS: Flute casts (downslope), normal grading, massive base**

4 Why NOT (d) Contour Current? - DETAILED EXPLANATION

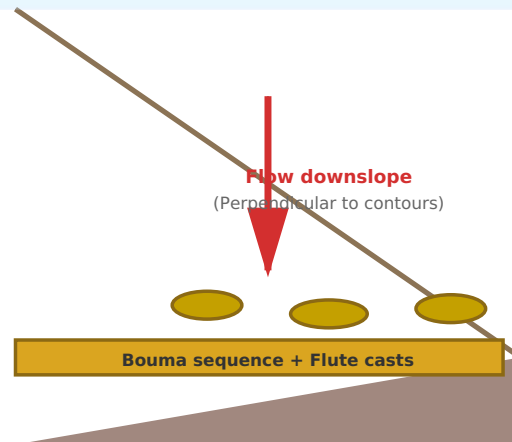
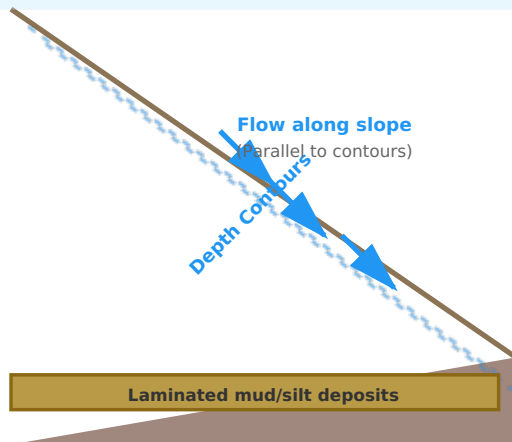
Contour Currents: Flow Direction and Deposits

Contour Current vs Turbidity Current

CONTOUR CURRENT

Ocean/Lake Water

TURBIDITY CURRENT



Contour currents flow **PARALLEL** to depth contours (sideways along slope) Turbidity currents flow **DOWNSLOPE** (perpendicular to contours)

Key Differences: Why Contour Current $\neq$ Your Bed

1. FLOW DIRECTION:

- **Contour currents:** Flow **PARALLEL** to continental margin, following depth contours (sideways along slope)
- **Your bed:** Has **FLUTE CASTS** pointing downslope $\rightarrow$ indicates **DOWNSLOPE** flow (perpendicular to contours)
- **✗** Contour currents **DON'T** create flute casts (which indicate downslope scour)

2. SEDIMENTARY STRUCTURES:

- **Contour currents:** Produce laminated muds and silts, with ripple marks, but **NOT** massive bases
- **Your bed:** Has organized Bouma sequence (massive $\rightarrow$ laminated $\rightarrow$ rippled)
- **✗** Contour currents don't produce the organized massive-to-rippled progression

3. GRADING PATTERNS:

- **Contour currents:** Deposit fine-grained material uniformly; no coarse-to-fine normal grading
- **Your bed:** Shows clear **NORMAL GRADING** (coarse at base, fine at top)
- **✗** Contour currents don't create the progressive grain-size reduction with organized structures

4. FLOW ENERGY & MECHANISM:

- **Contour currents:** Driven by density differences (thermohaline circulation), steady flow, **LOW** energy
- **Your bed:** Shows **HIGH INITIAL** energy (flute casts, massive) transitioning to **LOW** energy (ripples)
- **✗** Contour currents don't have the energy to create downslope scour marks

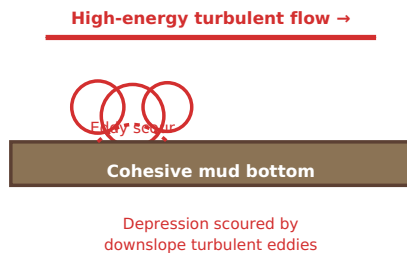
CONCLUSION: Your bed is DEFINITELY NOT a contour current deposit. It shows all the hallmarks of a HIGH-ENERGY downslope turbidity current!

5 Flute Casts: The Smoking Gun

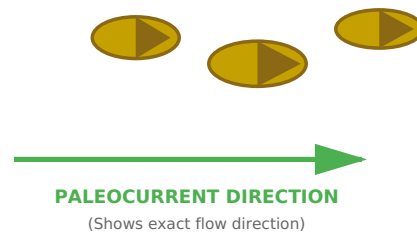
Why Flute Casts Rule Out Contour Currents

Flute Cast Formation & Paleocurrent Significance

Formation Requires DOWNSLOPE Flow



Result: Flute Cast



Contour Currents CANNOT Create Flute Casts

- ✗ Contour currents flow PARALLEL to slope (sideways), not DOWNSLOPE
- ✗ They lack the DOWNSLOPE velocity needed for eddy scour that creates flute depressions
- ✗ YOUR BED HAS FLUTE CASTS → Must be from DOWNSLOPE flow → TURBIDITY CURRENT, NOT CONTOUR CURRENT

Flute casts prove downslope scour - this is ONLY possible with turbidity currents, not contour currents

6 Complete Comparison Table

Feature	(a) Debris Flow ✗	(b) Grain Flow ✗	(d) Contour Current ✗	(c) Turbidity Current ✓
Flow Direction	Downslope, unidirectional	Downslope, gravity-driven	PARALLEL to slope (sideways)	DOWNSLOPE, high-energy ✓
Sole Marks (Flute Casts)	None typical	None	Ripple marks ONLY (no flute casts)	FLUTE CASTS ✓
Internal Structure	Chaotic, no layers	Massive, no layers	Laminated mud/silt	Massive → Laminated → Rippled ✓
Grading	Poor/Reverse	No/Reverse	NO grading	NORMAL grading ✓
Grain Size Progression	Random mixing	Uniform massive	Fine material only	Coarse → Fine (Bouma) ✓

Bouma Sequence	No	No	No	COMPLETE ✓
Your Bed Match	X NO	X NO	X NO (lacks flute casts, grading)	✓ PERFECT!

🎯 FINAL ANSWER



Answer: (c) TURBIDITY CURRENT

Evidence: Flute casts + Normal grading + Bouma sequence (Massive → Laminated → Rippled) = PERFECT TURBIDITE

📖 TEXTBOOK SOURCE:

Boggs, Sam Jr. (2006). *Principles of Sedimentology and Stratigraphy*, 4th Edition. Pearson Prentice Hall, Upper Saddle River, New Jersey. ISBN: 0-13-154728-3

Document Information:

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Original Question: DMG 2019

Complete solution with clear diagrams and detailed explanations for all four options

Perfect for: Geology courses, exams, and competitive exam preparation

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